

Vishal Dhanasekaran

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Education

Coimbatore Institute of Technology

October 2022 – Present

M.Sc., Software Systems (Integrated) | CGPA: 8.18 (till 6th Semester)

Geethanjalee Matric Hr Secondary School

May 2022

HSC | Percentage: 85.3%

Professional Experience

Software Development Engineer, Asthra Medtech Pvt Lmt

June 2025

Coimbatore, On-Site

– November 2025

Ultrasound Scan Report Generating Software

Tech Stack: Next.js, Express.js, Postman

- Developed a web-based platform that allows doctors to upload DICOM medical scans and automatically generate structured reports using OCR.
- Implemented a report editor with MPIN-secured digital signatures for secure publishing.
- Reduced report preparation time from 60 minutes to 5 minutes, significantly boosting doctors' productivity.

Project Management System

Tech Stack: Next.js, Express.js, MySQL, Docker, AWS RDS, Bruno

- Designed and developed RESTful APIs with JWT-based authentication for secure, role-based routing.
- Built features for live task tracking, schedule variance monitoring, and resource allocation, improving efficiency for both developers and project managers.

Projects

Melinia'26, | bun | hono | vite + react (typescript)

January 2025 – February 2026

- Architected and deployed a full-stack event management platform using Bun, Hono, React, Vite, and TypeScript in a monorepo architecture.
- Successfully handled 2000+ users and event registrations during the tech fest.
- Integrated Redis + BullMQ for asynchronous job processing, and designed PostgreSQL database schemas to support scalable event registrations, payment tracking, and transactional email workflows.
- Implemented production-grade infrastructure with Docker, Caddy reverse proxy, GitHub Actions CI/CD, and Prometheus + Grafana monitoring, deploying the entire system on a self-managed VPS with a self-hosted Mailcow SMTP server for reliable email delivery and system observability.

Chotta Chat, | Next.js | Ollama | Vercel's AI SDK

August 2025

- A lightweight chat application that enables seamless interaction with locally hosted AI models via Ollama.
- Features a clean, responsive UI with real-time streaming responses for an intuitive user experience.

FOSSStream, | React.js

July 2025

- A single-page application (SPA) that aggregates FOSS news from multiple blogging platforms.
- Implements infinite scrolling with smooth, efficient data fetching and a responsive interface for uninterrupted browsing.

DataDashBoardPro, | Data Mining | Statistical Analysis

April 2025

- Visualizes student participation trends across areas of interest, event types (workshops, meetups, competitions), modes (online/offline) & time periods (monthly/quarterly/yearly).
- Provides actionable insights to improve event planning and engagement.

TickTOS, | RTOS | System Programming | OS Development

December 2024

- A user-space simulation of a preemptive Real-Time Operating System (RTOS) with task scheduling, and real-time elevator control using priority-based scheduling.

CHESS GAME ENGINE, | ReactJS | Tailwind CSS

July 2024 – October 2024

- A chess engine, with the score sheet that appends the algebraic notations automatically.

ATTENDIX, | C Programming

March 2023 – May 2023

- Attendix is a terminal-based attendance management system entirely runs in C language.

Technical Skills

Programming Languages: C, C++, Python, bash, Java, asm x86, Typescript, Kotlin (basic)

Tools/Frameworks: Next.js, React.js, Node.js, Git/Github, Linux, Ollama, Cypress, Docker

Core Team Member, FOSS United Coimbatore	June 2025 – Present
Organized Meetups in promote and build FOSS Ecosystem in Coimbatore region.	
Club Advisor, FOSS-CIT	May 2025 – Present
- Delivered a Flash task in IndiaFOSS 2025, India’s Largest FOSS Conference. - Represented the Club in TOSSConf 2025, a state level Open Source Software Conference	
Secretary, FOSS-CIT	May 2024 – April 2025
- Led the club by coordinating tasks and ensured fairness in teamwork - Delivered sessions on GNU/Linux, DSA, and Version Control Systems. - Team Lead & Maintainer for Open Source Projects of FOSS-CIT	
Public Relations Manager, FOSS-CIT	May 2023 – May 2024
Collaborated with participants and managed promotional activities of the Club.	

Co-Curricular

Introduction to JavaScript Security (LFS184), The Linux Foundation	November 2025
DIPLOMA in Quantum Computing, QWorld Bronze 2024	October 2024
WINNER in Tech Research Presentation, Computing Research Day'24 - CIT	September 2024

Interests

Backend Development | Open Source Software Development | Operating Systems | DSA